Which hand?

Detecting left- vs right-hand nose touches from smartphone motion sensors

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Abstract

A person brings a smartphone from arm's length to the nose with the left or the right hand. We asked whether the phone's motion sensors alone can tell which hand it was, as a probability $P(\text{right})$ that the right hand moved the phone, for people the model has never seen. We built a web app that records the movement and collected 598 usable movements from 20 participants. We tested several kinds of models, always on a participant left out of training, and a one-dimensional convolutional neural network (1D CNN) performed best: for new participants we estimate that it identifies the hand with an accuracy of 88.5 % (95 % confidence interval 84.1–92.3 %), with an area under the ROC curve of 0.945. We also checked that the predicted probabilities can be taken at face value (calibration): some models were over-confident and needed a correcting function, while the network's outputs were already close to calibrated. Finally, combining several movements made with the same hand improves the prediction: in a test of this step, the accuracy rose from 89.2 % with one movement to 96.3 % with three.

1 Introduction

The question. A person holds a smartphone at arm's length and brings it to touch the nose, with the left or the right hand (Figure 1). The task, set by the principal investigator, is to predict which hand it was from the phone's motion sensors alone. A phone has two such sensors: the accelerometer, which measures the phone's acceleration and also feels gravity, and the gyroscope, which measures how fast the phone rotates. Our recording app is a web page, and a web page can read both sensors [1], on our participants' phones about 60 times per second. The answer should be a probability, $P(\text{right})$, that the right hand moved the phone, and it should be calibrated: of the movements given 80 %, about 80 % should really be right-hand movements. Above all, it must work for people the model has never seen, since that is how it will be used.

Why it is hard. People differ in arm length, speed and style, and the model sees no movement of the person it is tested on. Phones differ in their sensors and in how the browser reports them [2]. The sensors measure along the phone's own axes, which turn with the way the phone is held, so the same arm movement gives different raw signals when the phone is held upright or sideways. Most importantly, the two arms are mirror images of each other, so simple descriptions of a movement, such as its speed, its duration or the length of its path, are about the same for both hands.

Our approach in brief. Figure 2 shows the steps from one movement to an answer. We built a web app that guides a participant through many such movements, asking each time for the left or the right hand, and records the two sensors. Each recording is then cleaned and expressed

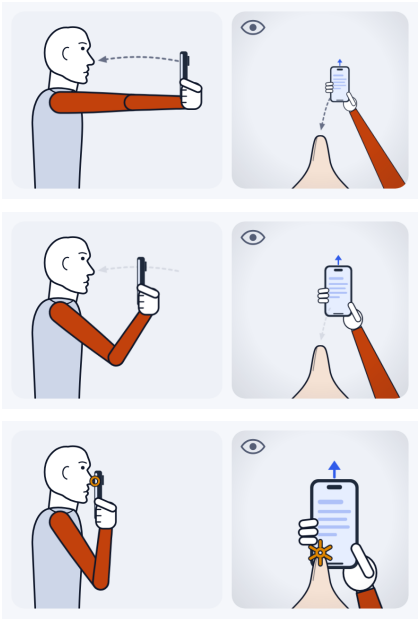


Figure 1: The movement as the app's instructions draw it (right hand, phone upright, natural speed): arm extended and still (top), on the way (middle), touch and hold (bottom); left of each frame the person seen from the side, right of it the phone as the person sees it. (The arm's colour is the app's; in our figures red marks the left hand and blue the right.)

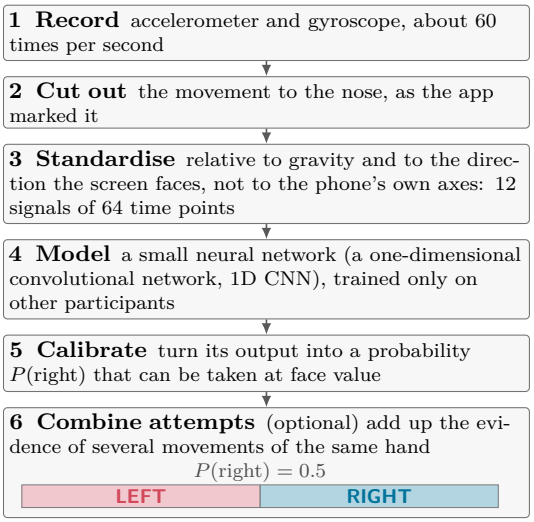


Figure 2: From a movement to an answer: the steps of the final pipeline. Section 2 explains step 1 and the app's markers, Section 3 steps 2 and 3, and Section 5 steps 4 to 6. Bottom bar: the answer, $P(\text{right})$, from 0 (left hand) to 1 (right hand); the middle line, 0.5, separates a left answer from a right one.

relative to gravity and to the direction the screen faces instead of the phone's own axes, so that it no longer depends on how the phone sits in the hand. Physics tells us where to look: because the arms are mirror images, the hand should show mainly in quantities that a mirror reverses, above all the direction in which the phone turns about the vertical. Early in the movement the phone should turn counter-clockwise (seen from above) in the right hand and clockwise in the left, and the recordings confirm this. Since this one quantity is not enough on its own, we trained several kinds of models, from a simple rule to a small neural network, and tested each one only on participants it had never seen. Finally, we checked that the predicted probabilities are calibrated, and we combined several movements made by the same person with the same hand into one answer.

Related work. Several studies have detected the hand that holds or operates a phone, or a user's dominant hand, mostly to adapt the screen to that hand. Goel et al.'s GripSense used the touchscreen, the gyroscope and the vibration motor to recognise how the phone is held (84.26 % of the hand postures recognised correctly, 10 participants) [3]. Löchtefeld et al. recognised the hand used for the unlock gesture from the phone's sensors during that gesture, with a nearest-neighbour classifier, which gives a new example the label of the most similar recorded one (98.51 % for 12 participants, all right-handed) [4]. Guo et al. recognised the operating hand from the finger's traces on the touchscreen, and changes of hand from the accelerometer and the gyroscope (about 94 % of the detections correct, 14 participants) [5]. Nelavelli and Ploetz used a single number that, like the turn of our phone, reverses between the hands: the direction in which a swipe on the screen

curves (99.53 % agreement with the hand, 12 users) [6]. Fernández et al. used touch measurements of 174 participants (a detection rate of up to 99.92 % after seven interactions) [7], Chen et al. the motion sensors with a neural network that reads the signal step by step (an LSTM; from 89.2 % in real life to 97.4 % while sitting) [8], and Singh and Singh the accelerometer, gyroscope and orientation sensor with a classifier built from many decision trees (XGBoost; 85.04 %, with three possible answers rather than two) [9]. These numbers are hard to compare with ours, because several of these evaluations let the same person appear in both the training and the test data: Guo et al. and Fernández et al. split their data at random, Nelavelli and Ploetz held out no users, and for Chen et al. and Singh and Singh it is unclear. A model can then recognise the person rather than the hand, which overstates the accuracy for new people [10, 11]: in a review of sensor and smartphone studies, the median reported error was 5.6 % with random splits and 13.0 % when whole participants were held out [12], and even the movement that brings a phone to the ear can identify its owner [13]. With few participants, an estimate also has wide error bars [14]. The closest work to ours is clinical: in the finger-to-nose test of a neurological examination, Mohammadi-Ghazi et al. saw with wrist motion sensors that the wrist rotates in opposite senses for the two hands (17 participants), but built no classifier [15]. Finally, modern neural networks are often over-confident [16], so a predicted probability must be checked for calibration.

2 Data

The app. We built the app that recorded the data, NoseTouch: a web app that runs in the phone's browser on iPhones (iOS) and Android phones, with nothing to install. It guides the participant hands-free with spoken and visual prompts, records the accelerometer and the gyroscope, and uploads every recorded movement. We also built a second page that runs the final model of this report in the phone's browser and shows which hand it predicts for a new person's movement (the inference page, `https://axelfridman.github.io/nosetouch-study/?predict=1`).

The protocol. A participant holds the phone still at arm's length, with the screen facing them. At a start beep they bring the phone to touch the nose, hold it there until a double beep, and stretch the arm out again (Figure 1); one such movement is a trial. The app asks for one hand at a time, in short blocks, and varies the grip (the phone upright, or sideways with its top to the left or to the right), the speed (natural, fast or slow) and, in some trials, the movement itself (for example a curved path or a pause halfway). While recording, the app detects when the phone starts to move after the beep and when it comes to rest at the nose (it needs this moment to play the double beep), and it stores these two moments with the recording; we call them the app's markers. The inference page has no start beep: it must find each movement in the sensor stream by itself, a step that this report does not evaluate (every accuracy in it uses the app's markers), so its answers may be less accurate. A participant's visit, one session with the app, took a median of 9.7 minutes.

Participants. The 20 participants, P01 to P20, were Bar-Ilan University students, most of them master's students, each offered sweets for their time (Figure 3; a video shows a session[1]). They sat or stood while recording (18 and 6 sessions), and 3 of them recorded a second or third session, so there are 24 sessions in all. They were 12 women and 8 men with a mean age of 25.1 (SD 2.9) years, from 19 to 29. The mean height was 164.2 (SD 6.5) cm for the women and 177.2 (SD 5.7) cm for the men (11 of the 12 women and 6 of the 8 men gave their height). They also reported their handedness (18 right-handed, 2 left-handed) and the hand they usually hold a

[1] A video of a data-collection session:
`https://drive.google.com/file/d/1V9PF623HkJ8MjR6loJhUHnDkCx60W-Jl/view?usp=sharing`.



Figure 3: A data-collection session (cropped so that no face is visible). Each participant follows the prompts on their own phone, starting from arm's length (the phone on the left asks for the left hand at natural speed); the sweets on the table were the compensation.

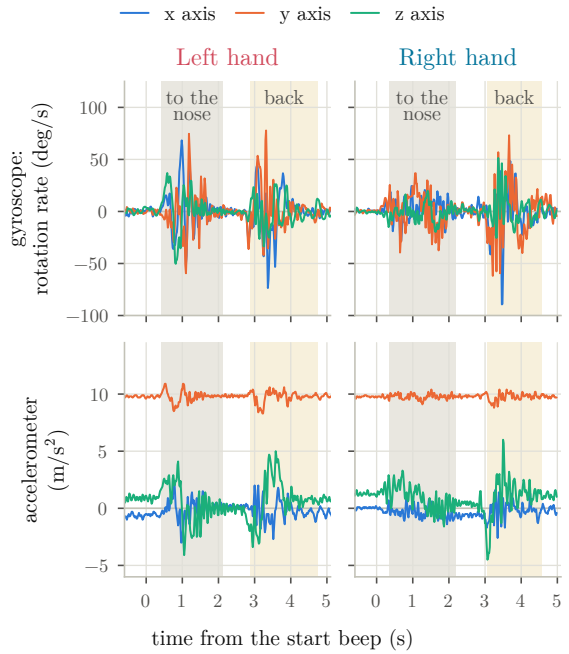


Figure 4: Two trials of P15, one with each hand (the same Android phone, upright grip, natural speed), exactly as the phone recorded them, about 60 samples per second with no smoothing (here the colours mark the axes, not the hands). Top: the gyroscope, the rotation speed about each of the phone's axes (x across the screen, y along it, z out of it); bottom: the accelerometer along the same axes, gravity included (in this grip the y axis points up, so it reads about 9.8 m/s^2 even at rest). Grey band: the movement to the nose, between the app's markers; beige band: the arm going back. Time runs from the start beep.

phone in; 12 recorded on an Android phone and 8 on an iPhone. One more person, PD01, who has a diagnosis of Parkinson's disease and is in their fifties, recorded with the same app. PD01 was never used to build or test any model and is described separately in Section 7.

What we kept. Of the 878 tries recorded (including those the app rejected), we kept 598 trials: those that the app accepted (among other checks, the phone started to move after the start beep and came to rest at the nose within the time limits, and the participant did not ask to redo the trial), that lasted at least 0.25 s, in which the phone clearly turned (a peak rotation speed of at least 30 degrees per second), and whose stream of samples had no gap longer than 100 ms around the movement (phones sometimes stop sending samples for a moment). We did not use a warm-up trial at the start of each visit, in which the participant chose the hand, nor a few discarded sessions: one that a participant stopped and later repeated, and the failed recordings of one more volunteer. The label of a trial is the hand the app asked for. The 598 trials are 302 left-hand and 296 right-hand trials, a median of 28 per participant (from 8 to 45).

What a recording looks like. Both sensors measure along the phone's own three axes: x across the screen (towards its right edge), y along the screen (towards its top) and z out of the screen. The gyroscope gives how fast the phone rotates about each axis, in degrees per second; the

accelerometer gives the acceleration along each axis, in m/s^2, and also feels gravity, so an Android phone held still and upright reads about 9.8 m/s^2 on its y axis (Section 3 explains iPhones). Figure 4 shows one left-hand and one right-hand trial of the same participant exactly as recorded: six noisy channels per trial (three rotation speeds and three accelerations), in which the two trials differ, but nothing tells the eye which hand moved the phone.

3 From raw signals to a standard movement

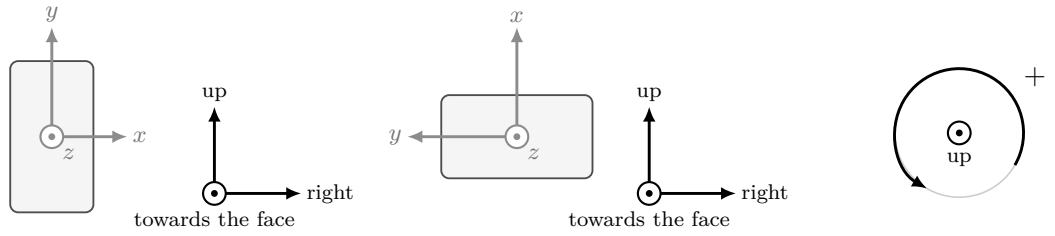
An even clock. A web page receives the motion samples about 60 times per second: per session (one phone on one visit), the median rate ranged from 59.9 Hz (on iOS) to 60.0 Hz (on Android). The samples do not arrive at perfectly even times, and some iPhone recordings have gaps of about a third of a second (66 recordings, median 318 ms; the lowest average rate of a single recording was 56.4 Hz), so we put every recording on an even 60 Hz clock by linear interpolation, bridging gaps of up to 100 ms (a trial with a longer gap near the movement was not kept, Section 2). This loses essentially nothing: the phones delivered about as many samples as the clock has, and the arm movement is slow (76 % of the gyroscope's signal power lies below 5 Hz).

Smoothing. We smooth every channel with a standard low-pass filter (a Butterworth filter of order 4, cut-off 15 Hz), run once forward and once backward in time so that it does not shift peaks in time. It is not a sharp cut: it damps fast wiggles gradually, keeping 98.8 % of the amplitude of a 10 Hz oscillation, 50.0 % at 15 Hz and 1.2 % at 20 Hz. During the movement, 99.2 % of the gyroscope's power and 98.4 % of the accelerometer's lie below 15 Hz, so the filter removes little of the movement itself.

Up, and the turn about the vertical. The phone's own axes turn with the grip, so we express the motion relative to the vertical. When the phone is still, the accelerometer feels only gravity, which shows which way is down; we measure it in the still 300 ms just before the movement (at arm's length) and just after it (at the nose). During the movement the accelerometer also feels the movement itself, so there we carry 'down' with the gyroscope: it tells how far the phone has rotated since the still moment, and we rotate 'down' back by the same amount, sample by sample, forward from the start and backward from the end, and blend the two. (We did not use the phone's compass, the magnetometer, so we know where up is but not which way the person faces; phones also report their own estimate of gravity, but we computed ours.) Carried through the whole movement, the 'down' of the start arrives within a median of 1.1° of the 'down' measured at the nose.

With up known at every sample, we subtract gravity from the accelerometer, so that every acceleration from here on is that of the movement alone (zero when the phone is still), and we split each rotation and acceleration into a vertical and a horizontal part. This gives 8 *gravity-frame* channels (a frame is a set of directions to measure along; this one is fixed by gravity alone) that do not depend on how the phone sits in the hand: the total rotation speed, the total acceleration, the vertical and the horizontal acceleration, the tilting speed (how fast the phone tilts about any horizontal axis, a size without a direction), the yaw rate, the yaw and the tilt since the start (how far the phone has tipped relative to the vertical since the start of the movement). The *yaw rate* is how fast the phone turns about the vertical, like a door on its hinges, counted positive counter-clockwise seen from above (Figure 5c); the *yaw* adds it up from the start of the movement, that is, how far the phone has turned about the vertical so far.

The iPhone sign. Measuring 'down' needs the right sign. iPhones report the accelerometer with the opposite sign to Android phones on all three axes (the gyroscope has the same sign on both),



(a) phone upright (b) phone sideways (top to the left) (c) seen from above

Figure 5: The phone's own axes (grey) and the directions we measure in (black; this section explains them), as the person sees the phone, so that z and 'towards the face' point out of the page. (a, b) Turning the phone in the hand from upright to sideways swaps its x and y axes, but up, right and towards the face stay the same. (c) Seen from above, up points out of the page: the yaw rate, the turn about the vertical, is positive counter-clockwise.

so on an iPhone 'down' would come out as 'up', every turn about the vertical would be reversed and a left hand would look like a right hand. We therefore multiplied every iPhone accelerometer value by -1 and checked each session on its own recordings: while the phone is held still before a movement, the corrected reading must point up along the axis that the grip puts on top. All 9 iPhone sessions needed the flip, and none of the 15 Android sessions did.

The body frame. The gravity frame has no horizontal direction, so it cannot tell which way the phone moves or tilts sideways; the phone's own axes can, but they change with the grip. Four more channels therefore use two horizontal directions set by where the screen faces (Figure 5): 'towards the face', the direction the screen faces, made horizontal, and 'right', horizontal and perpendicular to it, towards the person's right. Since the screen faces the person, these are in practice the directions of the person's body (hence the name). They turn with the phone when it turns about the vertical, but not when it is turned within the plane of its screen, which is all that a change of grip does: held upright or sideways, the screen still faces the person, so up, right and towards the face, and every channel measured along them, stay the same, while the phone's own x and y axes swap places (Figure 5a, b). Along these directions we measure the roll rate (rotation about the towards-the-face direction) and the pitch rate (rotation about the right direction), which split the tilting speed into its two horizontal directions, with signs, and the sideways acceleration and the acceleration towards the face. The final model reads these 4 *body-frame* channels together with the 8 gravity-frame channels.

Cutting out the movement, and time-normalising it. The models read only the movement to the nose, between the app's two markers. The start marker is the last still moment before the phone's rotation or acceleration rose clearly above its resting level, a median of 575 ms after the start beep; the end marker is the moment the phone came to rest at the nose. A median movement lasts 1.65 s, that is, 99 samples at 60 Hz. Between the two markers we compute each channel at 64 points evenly spaced in time, by interpolating between the samples, whatever the duration: the first point is the start and the last is the arrival at the nose. Time then runs as a percentage of the movement, from 0 to 100 %, so movements of different speeds line up point by point, and every movement becomes an input of the same size.

4 Why the hand is detectable

The mechanism. Reflecting a movement left to right, as in a mirror, gives the same movement made with the other hand. The reflection keeps the speed, the duration, every up–down and

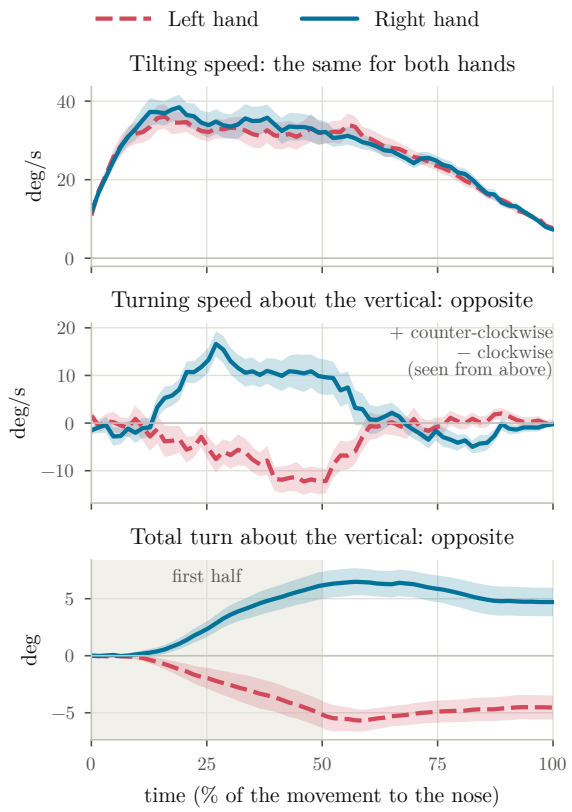


Figure 6: The average movement to the nose with each hand (left: red, dashed; right: blue, solid): each participant's mean curve, averaged over the 20 participants (band: ±1 standard error). Time runs from the start of the movement (0 %) to the arrival at the nose (100 %). Top: tilting speed; middle: yaw rate (turning speed about the vertical); bottom: the yaw, with the first half of the movement shaded.

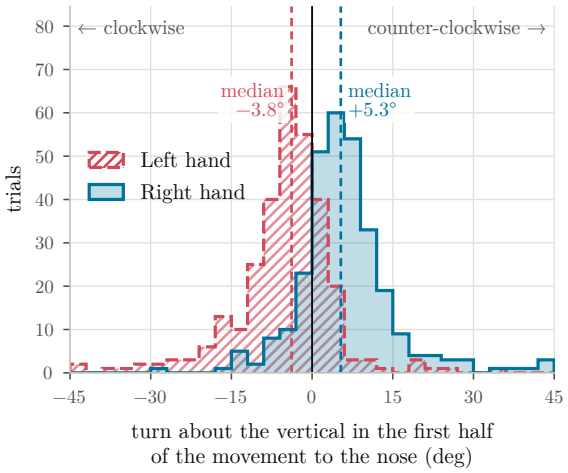


Figure 7: The first-half yaw of all 598 trials (302 left-hand, hatched; 296 right-hand, filled): how far the phone turned about the vertical in the first half of the movement to the nose (positive: counter-clockwise seen from above). Dashed: the medians of the two hands. Trials beyond ±45° are drawn at the edge.

towards-the-face motion and the tilting forward and back, but it reverses every sideways motion, every turn about the vertical (clockwise becomes counter-clockwise) and every roll, the turn about the towards-the-face direction. Because the two arms are close to mirror images, the hand can show mainly in such reversed quantities, and the turn about the vertical is the most natural one. When the phone is brought to the nose, the forearm swings towards the body's midline, so, seen from above, the phone turns counter-clockwise in the right hand and clockwise in the left. Near the nose the wrist turns the phone partly back to face the person, so the turn in the first half of the movement should carry the clearest signal.

The evidence. The average movements of the 20 participants show exactly this (Figure 6). The phone tilts at the same speed with both hands (top panel), but it turns about the vertical in opposite directions over most of the first half of the movement, counter-clockwise with the right hand and clockwise with the left (middle); added up, the yaw of the two hands ends with opposite signs (bottom). To summarise each trial by one number, we take its *first-half yaw*: the yaw at the halfway point in time, that is, the yaw rate added up from the start of the movement until halfway to the nose. Its median is −3.8° for the left hand and +5.3° for the right (Figure 7), and its sign alone (positive: right hand), with no training at all, names the hand in 79 % of the trials.

The two histograms overlap, however, so on the remaining trials the sign names the wrong hand: one number is not enough.

How well a number separates the hands is measured by the AUC, the area under the ROC curve (which traces the share of right-hand trials called right against the share of left-hand trials called right as the decision threshold moves). Its meaning is simple: the chance that a randomly chosen right-hand trial has a larger value than a randomly chosen left-hand trial; it is 0.5 for a useless number and 1 for a perfect one. For the first-half yaw the AUC is 0.85. Within each participant it is above 0.5 for all 20 participants (median 0.90), and it is about the same on Android phones (0.86) and on iPhones (0.84). The yaw at the end of the movement separates the hands less well (AUC 0.77), because the wrist turns the phone back near the nose. Numbers that a mirror leaves unchanged carry almost no information about the hand: of the 71 single-number summaries of the movement that we examined, the 14 that the mirror leaves unchanged, such as the peak rotation speed (AUC 0.52) and the duration (0.52), have AUCs between 0.46 and 0.53, while the 57 that the mirror reverses reach up to 0.87 (for example the yaw one second after the start of the movement), a little above the first-half yaw.

5 Models and how we tested them

Testing on new people. To measure how a model does on a person it has never seen, we used leave-one-subject-out (LOSO) cross-validation: a model is trained on 19 participants and tested on the remaining one, and this is repeated 20 times, so that every participant is the test participant once. Everything a model learns from data, including its calibration and the factor used to combine attempts (both described below), is learned from the 19 training participants only. Some of these steps need predictions for the training participants made by a model that was not trained on them (a model predicts the people it was trained on better than new people, so calibrating or combining on such predictions would make it over-confident). For these steps we split the training participants into a few groups, trained on all groups but one and predicted the group left out, in turn; we call this the *training-only split*. The predictions of the 20 LOSO rounds are then pooled and scored together.

Measures. The *accuracy* is the share of trials in which the hand given more than 50 % is the hand actually used. The AUC of Section 4 measures how well the predicted $P(\text{right})$ separates the hands. The *log loss*, or binary cross-entropy, is the average over the trials of $-\ln$ of the probability given to the hand actually used: it is low when the model is right and confident and punishes confident mistakes heavily; a model that always answers 50 % scores 0.693, and lower is better. The main numbers come with a 95 % confidence interval (CI), written [low, high]: unless stated otherwise, we resampled the 20 participants with replacement 2000 times, recomputed the number each time and kept the middle 95 % of these values (a person bootstrap, which respects that the trials of one person are alike).

The models. Because one number is not enough, we built models that read more of the movement, in three families: a physics model that uses a few numbers chosen from the mechanism of Section 4; standard machine learning on many summary numbers of each trial (gradient boosting and a stack of three models); and models that read the whole time series (MiniRocket and a 1D CNN). A simple yaw baseline completes the comparison. Every model reads the movement to the nose of Section 3 and outputs a score s, the log-odds of the right hand, $s = \ln\big(P(\text{right})/(1 - P(\text{right}))\big)$: $s = 0$ means 50 %, positive values favour the right hand, and $P(\text{right}) = \sigma(s) = 1/(1 + e^{-s})$.

- *Yaw baseline.* A logistic regression on one channel, the yaw, read at 8 evenly spaced moments of the movement. It was fixed before any other model was built, as the baseline that a more complex model has to beat.
- *Physics model.* An antisymmetric logistic regression on 6 numbers chosen from physics: the yaw, the roll (the turn about the towards-the-face direction so far) and the sideways velocity (the sideways acceleration added up), each read halfway through the movement and at its end. All six change sign under the mirror and the regression has no intercept, so the mirror flips the sign of s, and since $\sigma(-s) = 1 - \sigma(s)$ the mirror image of a movement gets exactly $1 - P(\text{right})$: the two hands are treated symmetrically by construction.
- *Gradient boosting* (LightGBM): 100 small decision trees, each correcting the errors of the previous ones. A decision tree sorts a trial by a few yes/no questions about its numbers (for example, is the average yaw in some time segment above a threshold?), and each final group, a leaf (at most 15 per tree), adds its own amount to the score. The trees read 238 summary numbers: for each of 17 channels (the 8 gravity-frame channels and, along each of the phone's three axes, the acceleration, the rotation speed and the direction of up, which shows how the phone is held), its average in 8 equal time segments, its mean, spread, minimum and maximum, and when these two occur.
- *Stack of 3 models.* A logistic regression, a support vector machine (a classifier that draws a curved boundary between the hands) and gradient boosting, each on 691 summary numbers per trial (averages in time segments, statistics, frequency content, correlations between channels and running sums, among others), combined by a small logistic regression that learns how much to trust each, from their predictions in the training-only split.
- *MiniRocket* [17], a fast general-purpose method for time series. It slides a fixed set of short patterns along the 8 gravity-frame channels: at each moment a pattern is multiplied point by point with the stretch of signal under it and the products are added up, which gives a large response where the signal looks like the pattern. Each pattern is also stretched to several time scales, and for each the share of the movement where it responds becomes one summary number (3948 in all); a logistic regression is fitted on these numbers.
- *1D CNN*, the final model: a small one-dimensional convolutional neural network that reads the 12 gravity- and body-frame channels as time series of 64 points. Like MiniRocket it slides short patterns along the signals, but it learns them from the training data, and it stacks them in 3 layers of 32 patterns, 7, 5 and 3 points long: the first layer compares its patterns with the channels, and each later layer looks for patterns in the responses of the layer before, so that it finds longer patterns built from shorter ones. The responses are averaged within 8 time segments, so that the network keeps when in the movement a pattern occurred, and a weighted sum of these averages gives s. The numbers in the patterns and in the sum, the weights (11329 per network), are learned in many small steps that lower the log loss on the training trials. We split the training participants into 4 groups and trained 4 such networks, each on all groups but one, stopping when its log loss on the group left out no longer improved (which makes its predictions for that group slightly flattering); for a new participant their outputs are averaged. Training also used small random stretches in time, changes of scale and noise.

Mirror images as extra training data. Every model but the physics model, which is symmetric by construction, was also trained on the mirror image of every training movement, labelled with the other hand (mirror augmentation): the same channels with every sideways motion, every turn about the vertical and every roll reversed (Section 4). This doubles the training data, balances the two hands and teaches the model that the two hands move as mirror images.

Calibration. A model's output need not be a calibrated probability: many models are over-confident, with probabilities too close to 0 and 1 [16]. We check calibration with a reliability curve: the trials are sorted by their predicted $P(\text{right})$ and cut into 10 groups of 59 or 60 trials,

and for each group the average predicted $P(\text{right})$ is plotted against the share of its trials that really were right-hand trials; a calibrated model lies on the diagonal. A calibration function can correct a model that is not calibrated. We used Platt scaling [18]: the calibrated probability is $\sigma(as + b)$, where s is the model's score; $a < 1$ pulls extreme scores towards 50 %, which corrects over-confidence, $a > 1$ pushes them away from 50 %, which corrects under-confidence, and b shifts all scores towards one hand. In each LOSO round, a and b are fitted on the training participants' predictions from the training-only split, so the left-out participant never informs its own calibration.

Combining several attempts. A person can repeat the movement with the same hand, and each such attempt (one more kept trial of the same person and hand) adds evidence about the hand. On the log-odds scale the evidence of independent observations simply adds up: two independent answers of 80 % each are odds of 4 to 1 each, together $4 \times 4 = 16$ to 1, that is 94 %, and multiplying odds is adding log-odds. The attempts of one person are not independent, however (same person, same habits), so adding them as if they were would make the combined answer over-confident. We therefore add the scores of the k attempts, divide the sum by a factor τ_k, and turn the result back into a probability:

$$P(\text{right})_k = \sigma\left(\frac{1}{\tau_k}\sum_{i=1}^{k} s_i\right), \tag{1}$$

where s_i is the score of attempt i. The factor τ_k lies between 1 (attempts fully independent: a plain sum) and k (attempts identical: a plain average of the scores). In each LOSO round it is the value that gives the lowest log loss on random groups of k attempts of the training participants, scored with their predictions from the training-only split. For three attempts its median over the rounds is 1.54: dividing the sum of three scores by 1.54 counts them as 1.9 independent attempts instead of 3, so three attempts are worth about two independent ones.

How we chose the processing. For the processing steps of Section 3, except the iPhone sign (a correction, not a choice) and the body frame (which only the CNN reads), we also tried alternatives, re-running the whole LOSO evaluation each time with two fixed test models (a logistic regression on the yaw rate and the yaw, similar to the yaw baseline, and gradient boosting) and, for several alternatives, also with MiniRocket: no smoothing or another filter, a coarser clock, the phone's own estimate of gravity, a window of fixed length instead of the time-normalised movement, channels rescaled to a common size, other sets of channels and no mirror images, among others (11 choices in all). No alternative was clearly better. In accuracy, two were clearly worse: replacing the yaw rate and the yaw by the signals along the phone's own axes changed the accuracy of the logistic regression by −12.5 [−17.4, −7.9] percentage points, and keeping only the gravity-frame channels changed that of gradient boosting by −6.0 [−8.8, −3.2] points (the phone's own axes still carry some information for this model; the final model nevertheless reads only channels that do not change with the grip, Section 3). By log loss, a few more were clearly worse for gradient boosting, among them training without the mirror images (a change of +0.057 [+0.027, +0.089]). All other differences were of a size that chance alone can produce. We therefore kept the processing described above.

Choosing the final model, and an honest estimate. We compared 17 models in all (several versions within each family) and kept the 1D CNN, which had the lowest log loss and the highest AUC. Picking the best of many models on the same participants makes the winner's own score slightly flattering, because part of its lead can be luck. To get an honest number for a new participant we repeated the choice inside every LOSO round: with one participant left out, the 17 models were compared by their log loss on the other 19 participants only (with the training-only split, in 5 groups), and the one with the lowest log loss there was tested on the left-out participant.

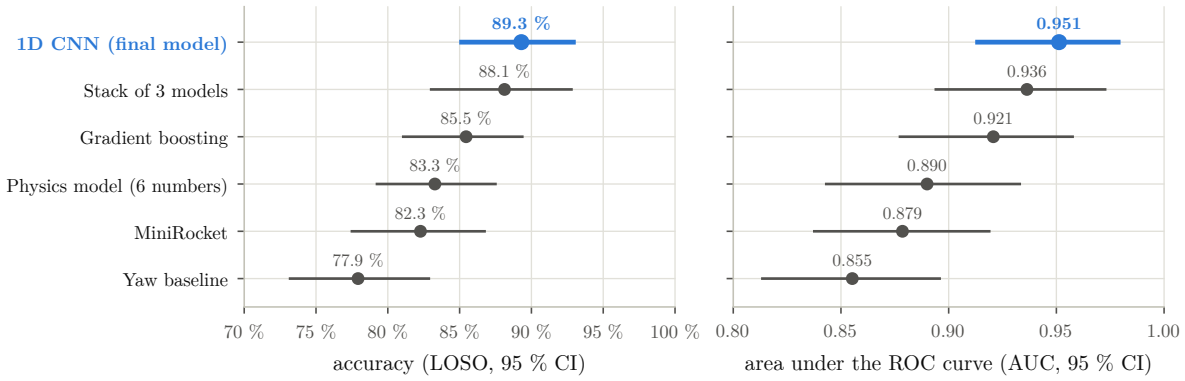


Figure 8: One version of each of the six models of Section 5 (17 models were compared in all), each tested on participants it had never seen (LOSO): the yaw baseline, the physics model, two models on summary numbers (gradient boosting and the stack) and two that read the whole time series (MiniRocket and the 1D CNN). Accuracy (left) and AUC (right) with their 95 % CIs. The 1D CNN, the final model (blue), scores highest on both.

In a round the choice can fall on another model, so this estimate measures our whole way of choosing a model, applied to a new person; on all 20 participants the same choice gives the CNN. This honest estimate is our main result. It repeats the final choice among the models, but not the earlier design decisions made while looking at the same results (for example, which channels the CNN reads), so it may still be slightly optimistic.

6 Results

Which model is best. Every model separates the hands far better than chance, and every other model scores higher than the yaw baseline (Figure 8). The yaw baseline itself does no better than the sign of the first-half yaw alone (79 %, Section 4): reading the yaw at more moments adds little, and the models that do better also read other channels. The 1D CNN performed best: in its own LOSO run (the slightly flattering score of Section 5) its accuracy is 89.3 % [85.0, 93.1] and its AUC 0.951 [0.912, 0.980], against 77.9 % [73.1, 82.9] and 0.855 [0.813, 0.896] for the yaw baseline. Its lead over the next models is small: compared with it on the same trials, participant by participant, the stack and gradient boosting could not be told apart from it, while MiniRocket, the physics model and the yaw baseline did detectably worse. Repeating the choice of the model inside every LOSO round gives the honest estimate for a new participant: an accuracy of 88.5 % [84.1, 92.3], an AUC of 0.945 [0.906, 0.975] and a log loss of 0.307 [0.212, 0.432].

Calibration. Gradient boosting, for example, was over-confident: groups of trials with an uncalibrated $P(\text{right})$ near 0 held more right-hand trials than claimed, and groups near 1 fewer (Figure 9, top left). One number summarises this pattern, the *calibration slope*: the a of a single Platt function fitted to all the pooled scores at once (1: nothing to correct; below 1: over-confident). For gradient boosting it is 0.76 [0.61, 0.99]. Its 20 fitted functions, one per LOSO round (a from 0.69 to 0.79), pull the scores towards 50 % (top middle). Afterwards the groups scatter on both sides of the diagonal instead of lying above it at low scores and below it at high ones (top right), and the slope is 1.00; the log loss changes little (0.369 before, 0.359 after). The final CNN's raw outputs were already close to calibrated (bottom left; slope 0.94): its fitted functions stay near the diagonal (a from 0.84 to 1.24), change its $P(\text{right})$ by 1.6 percentage points on average and leave its log loss practically unchanged (0.290 before, 0.291 after). We kept this step, planned for this model before the results, and the CNN's numbers in this report include it.

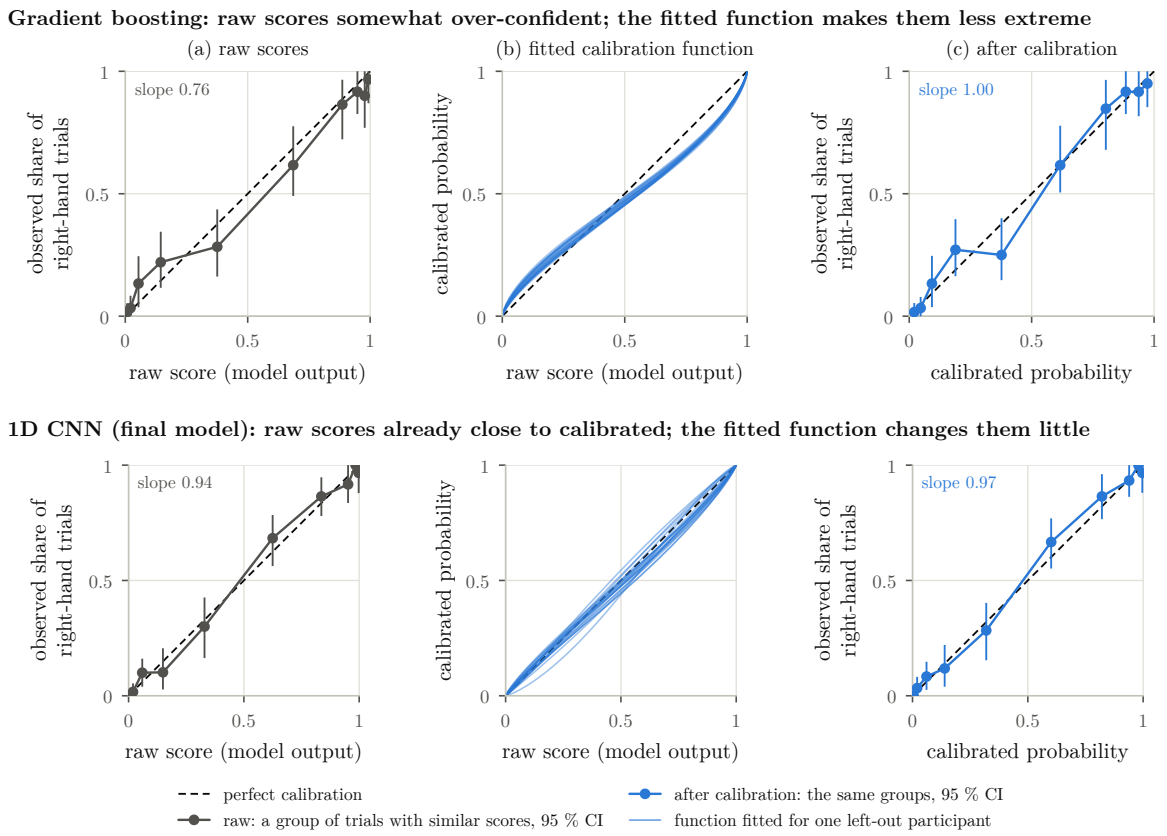


Figure 9: Calibration before and after, for a model that needed it (top: gradient boosting) and for the final model (bottom: the 1D CNN). (a) Raw scores (the uncalibrated $P(\text{right}) = \sigma(s)$) against the observed share of right-hand trials, in 10 groups of trials with similar scores (bars: 95 % CIs), with the calibration slope; the dashed diagonal is perfect calibration. (b) The fitted calibration function, one curve per LOSO round, each fitted on that round's training participants. (c) The calibrated probabilities against the observed share, with the calibration slope.

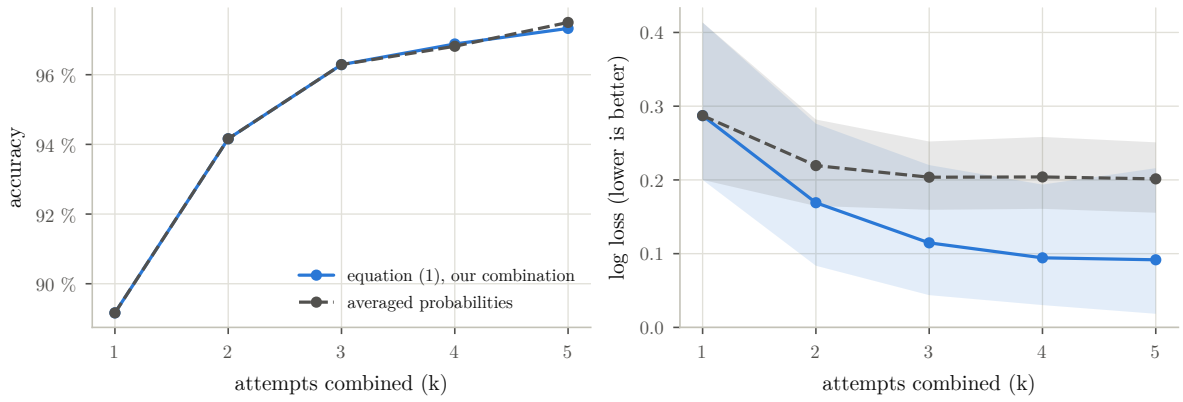


Figure 10: Combining k attempts of the same participant and hand (the 19 participants with at least 5 attempts of each hand; random draws of k attempts, each participant left out in turn; accuracy averaged over participants and hands): accuracy (left) and log loss (right, with 95 % bands), for the combination of equation (1) (blue, solid) and for a plain average of the probabilities (grey, dashed).

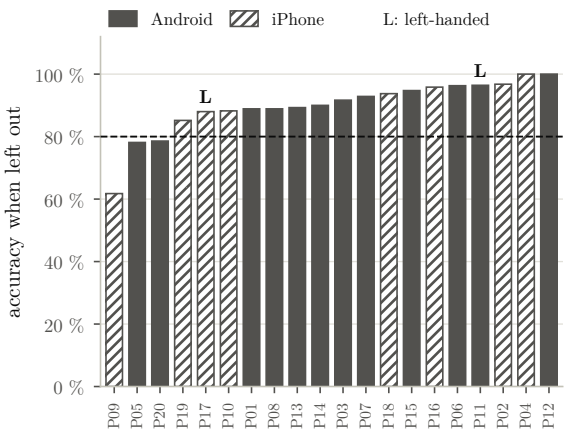


Figure 11: Accuracy of the final model for each participant when that participant was left out, sorted; dark bars: Android, hatched: iPhone; L: the two left-handed participants; dashed: 80 %.

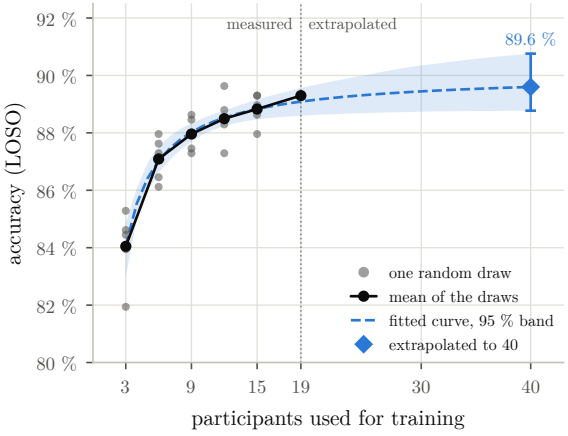


Figure 12: Accuracy of the 1D CNN trained on fewer participants (grey: one random draw; black: the mean of the draws), the fitted curve with its 95 % band, and its extrapolation to 40 participants.

Several attempts. Combining attempts with equation (1) helps (Figure 10). This test uses the 19 participants with at least 5 trials of each hand and averages the accuracy over participants and hands, so a single attempt scores 89.2 % here rather than the 89.3 % above. The accuracy rises from 89.2 % with one attempt to 96.3 % with three and 97.3 % with five, and the log loss falls from 0.287 to 0.115 and 0.092. A plain average of the probabilities of the attempts gives nearly the same answers, but its log loss levels off near 0.204 from three attempts on: an average does not become more confident as the evidence accumulates. Like the CNN's own score above, these numbers are slightly flattering, because the CNN was chosen on the same participants.

Who it fails on. The accuracy differs between participants (Figure 11). The median participant scores 90.8 %, 2 participants have every trial right, and 3 fall below 80 %, the lowest being P09 at 61.8 % (34 trials). The two left-handed participants score 96.4 % (P11) and 88.0 % (P17). The accuracy is 90.6 % on Android phones and 87.0 % on iPhones.

Would more participants help? We retrained the CNN inside every LOSO round on random subsets of 3, 6, 9, 12 and 15 of the training participants, drawing each size 5 times (Figure 12). The mean accuracy rises from 84.0 % with 3 training participants to 89.3 % with all 19, and flattens. We fitted a curve that rises quickly and then levels off towards a ceiling A, the inverse power law $\text{accuracy}(n) = A - B\,n^{-C}$ for n training participants (A = 90.1 %, B = 17.6 percentage points, C = 0.98), and read it at $n = 40$: 89.6 % [88.8, 90.8], only +0.5 points above the curve's own value at 19 participants (89.1 %). The interval, from refitting the curve 1000 times on resampled draws, understates the uncertainty, because all draws come from the same 20 people and the curve is extended beyond the measured sizes. More participants of the same kind would therefore add little accuracy; new ones would matter mainly to check the accuracy in groups we barely sampled, such as left-handed or older people and other phones, and for other kinds of movement.

7 Demographics, and a case study: PD01

Would demographics help? The app does not know who is using it. We asked whether knowing a participant's age, sex, height, handedness and usual phone hand would help if it did. In each LOSO round, a small second logistic regression combined the final model's score with these five

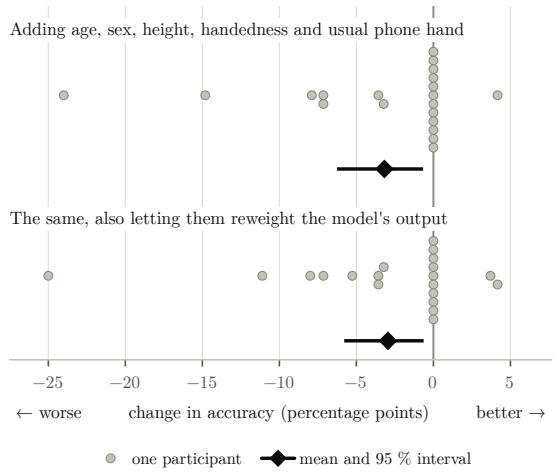


Figure 13: Change in accuracy of each participant when age, sex, height, handedness and usual phone hand are combined with the final model's output (top), and when each item may also make that output count more or less (bottom), against the same combination without them (LOSO). Diamond and bar: the mean change and its 95 % CI.

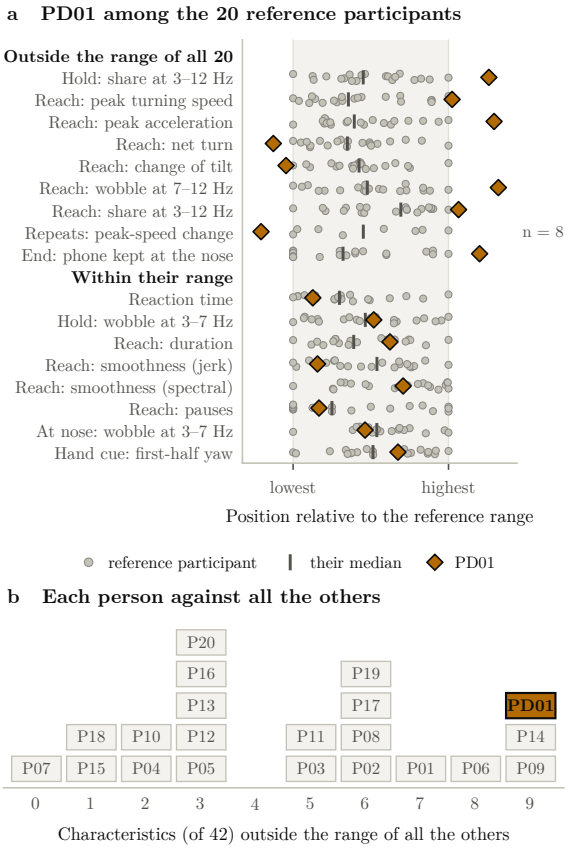


Figure 14: PD01 against the 20 participants (the reference participants). (a) One row per characteristic, rescaled from the lowest participant to the highest (shaded: their range; dots: the participants; tick: their median; diamond: PD01). Top: the characteristics in which PD01 lies outside the range of all the participants compared; bottom: characteristics in which a difference might be expected. Hold: before the movement; reach: the movement to the nose; at nose and end: after the arrival and when the recording stopped; repeats: over successive trials (n = 8: only the participants with enough repeats); wobble: rotation back and forth at the given frequencies; share at 3–12 Hz: the part of the rotation at these frequencies; net turn: the overall change of the phone's orientation over the reach; smoothness (jerk, spectral): two standard measures of how smooth the reach was; reaction time: from the start beep to the start of the movement. (b) For each person, the number of the 42 characteristics in which that person lies outside the range of all the others, each judged on as many trials as PD01 made.

items, and we compared it with the same combination without them. They did not help; they made the accuracy slightly worse (Figure 13): it stayed the same for 12 of the 20 participants, fell for 7 and rose for 1, a mean change of −3.2 [−6.3, −0.7] percentage points (−3.0 [−5.8, −0.6] when each item could also make the model's output count more or less), probably because with only 19 training participants the combination fits chance differences between people that do not hold for a new person. On their own, the five items predict the hand at chance level (50.3 %), because every participant used both hands: a fixed personal trait cannot tell which hand made a given movement.

PD01: who, and what was recorded. Here we ask how one person with Parkinson's disease moved in this task, compared with the 20 participants (the reference participants of Figure 14). No model was trained, tuned or tested on PD01, and nothing here is a diagnosis. PD01, introduced in Section 2, is decades older than the participants and recorded with the same app, remotely, seated and without supervision, in 2 short sessions (4.2 and 2.6 minutes, against 4.3 to 21.1 for the participants) that covered only the start of the protocol, with the phone upright. The app accepted 7 of PD01's 10 tries: 2 warm-up trials, not used (as for the participants), and 5 instructed movements, all with the left hand and the phone upright, which are compared below (3 of them at natural speed, which the measures of the reach use).

What differs. Before looking at PD01's data we fixed 42 characteristics of the movement: how still the phone was held in the 1.0 s before the movement (the hold); how fast, long, far and smoothly it moved to the nose (the reach); how much the phone wobbled, that is, rotated back and forth, at 3–7 Hz (the band of the tremor typical of Parkinson's disease at rest) and at 7–12 Hz; how the reach changed over repeats; what happened in the 0.6 s at the nose; and the first-half yaw. Each was compared between PD01's average and each participant's average over the same kind of trial. PD01 lies outside the range of all the participants compared (all 20, or the 8 with enough repeats for the change over repeats) in 9 of the 42 characteristics (Figure 14a, top). PD01's reach had a much higher peak acceleration, 3.8 times the participants' median (15.9 against 4.2 m/s^2; highest participant 10.3), and its wobble at 7–12 Hz was 3.0 times the median (44.6 against 15.1 degrees per second). Over the reach, the phone's tilt changed less than for any participant (2.9° against a median of 9.3°), and in all 5 trials PD01 had not yet lowered the phone when the recording stopped. While holding the phone before the movement, PD01 swayed unusually little at slow frequencies, which made the share of the hold's rotation at 3–12 Hz the highest of all, without more fast wobble than the participants. These 5 differences survive every check we made, for example judging each participant on only as many trials as PD01 made, or leaving out the trials in which PD01's accelerometer may have reached its limit (an accelerometer cannot report values above a maximum, so a stronger acceleration is cut off there). The others do not: the peak turning speed, just above the highest participant's, the fall of the peak speed over the repeats, the reach's net turn and the share of the reach's rotation at 3–12 Hz. Where a difference might be expected, PD01 was within the participants' range (Figure 14a, bottom): the 3–7 Hz wobble while holding (55th percentile), the duration of the reach (70th), its smoothness and pauses, and the hand cue that the models read, the first-half yaw (55th).

Unusual, but not unique. With 20 participants to compare with, even an ordinary person lies outside everyone else's range on a few characteristics by chance alone. We therefore asked every participant the same question: in how many of the 42 characteristics does that participant lie outside the range of the other 19, judged on as many trials as PD01 made? The answer is a median of 4.0, from 0 to 9, and 2 participants (P09, P14) reach PD01's count of 9 (Figure 14b). We also tested each characteristic with the Crawford–Howell test, a t-test made for comparing one person with a small group, corrected for testing 42 at once (which makes some look extreme by chance): the smallest p-value, 0.008 for the peak acceleration, became 0.18 after the correction, so no difference is statistically significant on the scales we fixed in advance (on a plain instead of a logarithmic scale, the peak acceleration and the reach's 7–12 Hz wobble would be). With one person against only 20 others such a test has little power, so this is weak evidence either way. PD01 is therefore among the most unusual people we recorded, but not unique. Moreover, PD01's age and phone, whether the arm rested on something while PD01 sat (no one watched the remote recording), and an accelerometer that may have reached its limit in 4 of the 7 accepted trials could explain the differences as well as the disease; telling these apart would need several patients and age-matched controls.

8 Discussion

It was possible to predict the hand that brought a phone to the nose from the phone's motion sensors alone in most cases, for people the model had never seen: we estimate that a small neural network identifies the hand of a single movement with an accuracy of 88.5 %, and of three movements made with the same hand with about 96.3 % (slightly optimistic, Section 6). The cue is physical: early in the movement the phone turns about the vertical in opposite directions for the two hands. We also explored ways to make the prediction more stable and valid, such as checking and correcting its calibration and combining several attempts. In one person with Parkinson's disease, several characteristics of the movement were unusual, but this hand cue was typical (55th percentile). The main limitation is the sample: only 20 participants, all students aged 19 to 29, of whom 2 are left-handed and 8 used an iPhone, so the accuracy may differ for older or left-handed people and for other phones.

References

[1] W3C. *DeviceOrientation Event Specification*. World Wide Web Consortium. Feb. 12, 2025. URL: `https://www.w3.org/TR/orientation-event/`.

[2] A. Stisen et al. "Smart Devices are Different: Assessing and Mitigating Mobile Sensing Heterogeneities for Activity Recognition". In: *Proceedings of the 13th ACM Conference on Embedded Networked Sensor Systems (SenSys '15)*. 2015, pp. 127–140.

[3] M. Goel, J. Wobbrock, and S. Patel. "GripSense: Using Built-in Sensors to Detect Hand Posture and Pressure on Commodity Mobile Phones". In: *Proceedings of the 25th Annual ACM Symposium on User Interface Software and Technology (UIST '12)*. 2012, pp. 545–554.

[4] M. Löchtefeld et al. "Detecting Users Handedness for Ergonomic Adaptation of Mobile User Interfaces". In: *Proceedings of the 14th International Conference on Mobile and Ubiquitous Multimedia (MUM '15)*. 2015.

[5] H. Guo et al. "Recognizing the Operating Hand and the Hand-Changing Process for User Interface Adjustment on Smartphones". In: *Sensors* 16.8 (2016), p. 1314.

[6] K. Nelavelli and T. Ploetz. *Adaptive App Design by Detecting Handedness*. 2018. arXiv: `1805.08367 [cs.HC]`.

[7] C. Fernández et al. "Implicit Detection of User Handedness in Touchscreen Devices through Interaction Analysis". In: *PeerJ Computer Science* 7 (2021), e487.

[8] T. Chen, K. Zhu, and M. C. Yang. "Deep-Learning-Based Unobtrusive Handedness Prediction for One-Handed Smartphone Interaction". In: *Multimedia Tools and Applications* 82.4 (2023), pp. 4941–4964.

[9] P. Singh and S. Singh. "Dominant Hand Detection Using Base Smartphone Sensors". In: *Innovations in Software Engineering Conference (ISEC 2025)* (NIT Kurukshetra, India). 2025.

[10] M. A. Little et al. "Using and Understanding Cross-Validation Strategies. Perspectives on Saeb et al." In: *GigaScience* 6.5 (2017), pp. 1–6.

[11] E. Chaibub Neto et al. "Detecting the Impact of Subject Characteristics on Machine Learning-Based Diagnostic Applications". In: *npj Digital Medicine* 2 (2019), p. 99.

[12] S. Saeb et al. "The Need to Approximate the Use-Case in Clinical Machine Learning". In: *GigaScience* 6.5 (2017), pp. 1–9.

[13] M. Conti, I. Zachia-Zlatea, and B. Crispo. "Mind How You Answer Me! Transparently Authenticating the User of a Smartphone when Answering or Placing a Call". In: *Proceedings of the 6th ACM Symposium on Information, Computer and Communications Security (ASIACCS '11)*. 2011, pp. 249–259.

[14] G. Varoquaux. "Cross-Validation Failure: Small Sample Sizes Lead to Large Error Bars". In: *NeuroImage* 180 (2018), pp. 68–77.

[15] R. Mohammadi-Ghazi et al. "Objective Assessment of Upper-Extremity Motor Functions in Spinocerebellar Ataxia Using Wearable Sensors". In: *Sensors* 22.20 (2022), p. 7993.

[16] C. Guo et al. "On Calibration of Modern Neural Networks". In: *Proceedings of the 34th International Conference on Machine Learning (ICML)*. Vol. 70. Proceedings of Machine Learning Research. 2017, pp. 1321–1330. arXiv: `1706.04599`.

[17] A. Dempster, D. F. Schmidt, and G. I. Webb. "MiniRocket: A Very Fast (Almost) Deterministic Transform for Time Series Classification". In: *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '21)*. 2021, pp. 248–257.

[18] J. C. Platt. "Probabilistic Outputs for Support Vector Machines and Comparisons to Regularized Likelihood Methods". In: *Advances in Large Margin Classifiers*. Ed. by A. J. Smola et al. MIT Press, 2000, pp. 61–74.